

Waqar Ahmed

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Professional Summary

Computer Engineering graduate from NUST EME College (May 2026) with a strong foundation in Quality Assurance (QA/SQA) and hands-on machine learning experience through an award-winning Final Year Project. Skilled in API testing (Postman), test case design, and SDLC/STLC methodologies, combined with practical experience building and evaluating machine learning pipelines. Passionate about delivering reliable, well-tested software and applying AI to solve real-world problems.

Education

B.E. Computer Engineering — NUST EME College, Islamabad	<i>Sep 2022 – May 2026</i>
ICS (Pre-Engineering) — Military College Sui, Balochistan	<i>2017 – 2019</i>
Matriculation — Military College Sui, Balochistan	<i>2015 – 2017</i>

Technical Skills

QA & Testing	Postman, JIRA, Selenium, SDLC, STLC, Test Case Design, Regression Testing, API Testing, SQL
Machine Learning	TensorFlow, Keras, Scikit-learn, NumPy, Transformer Models, Feature Extraction
Embedded Systems	8086 Microprocessor, Proteus, Verilog HDL, EDA Playground
Networking	Cisco (IP Addressing, Subnetting, Routing/Switching, VLANs, TCP/IP)
Programming	Python, C/C++, Verilog, MIPS Assembly
Tools	Git, VS Code, Unity/C#

Final Year Project

BCI-Based P300 Speller for Communication Assistance (Transformer-Based EEG Classifier) *Sep 2025 – May 2026*
Python, PyTorch, Signal Processing, LLM Integration

- Designed a Transformer-based P300 classification model achieving **96.9% accuracy** with 230k parameters, trained on the BigP3BCI dataset.
- Integrated Llama 3.1 70B via Groq Cloud API for LLM-based sentence prediction, improving communication speed for users with limited mobility.
- Developed in partnership with HM Hospital Rawalpindi's Neurology Department, under supervisor Dr. Soiyba Javed.
- Filed for patent; work recognized at COMPAC 2025.

Technical Projects

Hardware Implementation of a 32-Bit Pipelined MIPS Processor *May 2023*
Verilog, FPGA

- Designed and synthesised a modular 5-stage pipelined MIPS processor with runtime-configurable data forwarding and hazard detection units, under strict FPGA resource constraints.

RISC-V Modular Signal Processing Engine *Jun 2025*
MIPS Assembly, MARS

- Implemented FIR filtering, auto/cross-correlation, energy calculation, and zero-crossing rate modules in a modular processor.

T-Rex Game on FPGA

Dec 2025

Verilog, FPGA

- Implemented a classic runner game with real-time obstacles and display control in Verilog HDL.

Multi-Level Car Racing Game

Jan 2024

Unity, Blender

- Developed a multi-level car racing game using Unity Engine and Blender, applying 3D modelling and game design principles.

Certifications & Additional Courses

AI & Machine Learning

HEC Generative AI Training Certificate — Pak Angels

Applied Machine Learning Systems with FastAPI for Developers

Jun 2026

QA & Testing

API Testing a Real Web Application via Postman

Jul 2026

Networking

Cisco Networking Certification (CCNA-level: IP Addressing, Subnetting, Routing/Switching, VLANs, TCP/IP)

Design

UI/UX Design Certification — Coursera